



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Resolve the SA fuel outlook, then test Vopak's accessible market

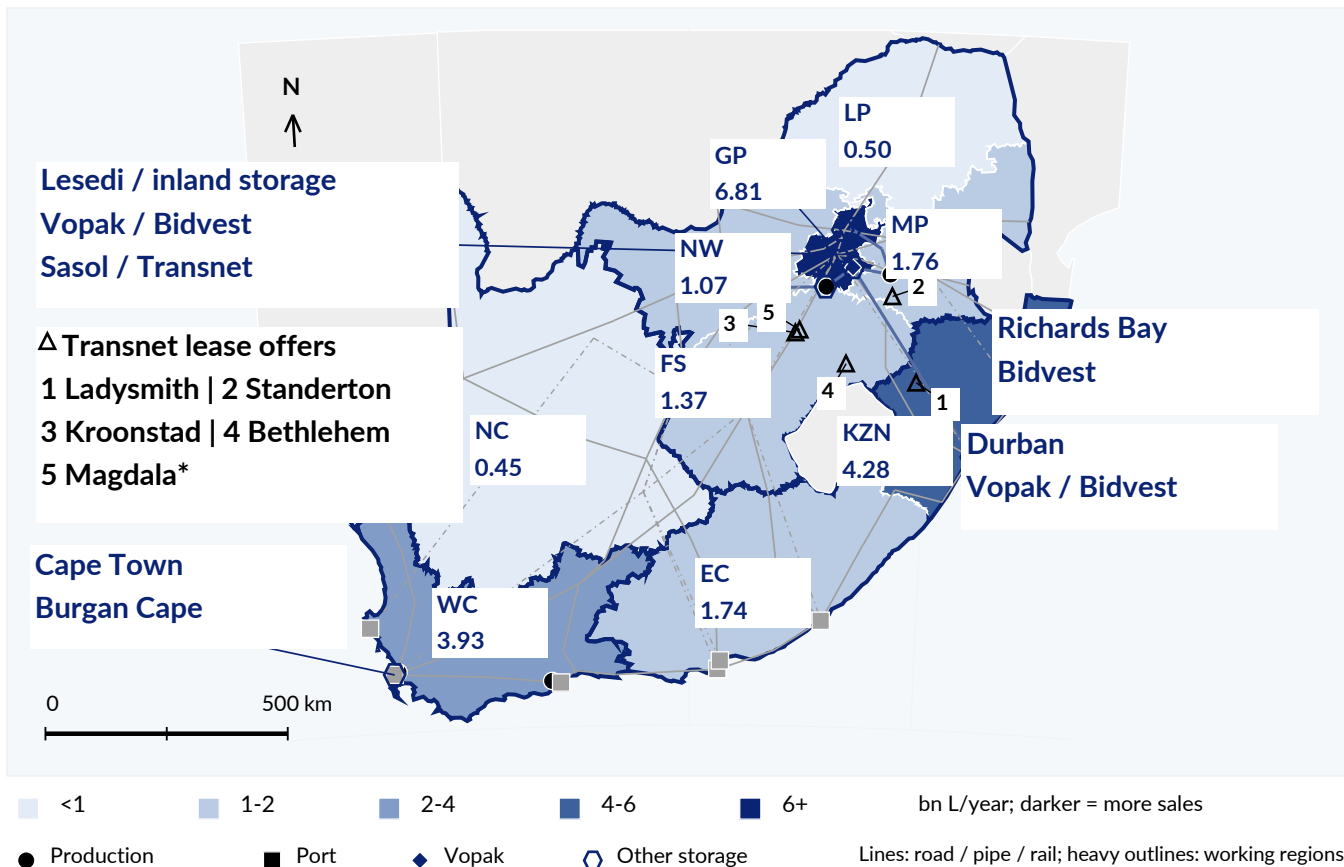
Situation → Complication → Resolution | evidence, actions and delivery roadmap

SCR reference	Evidence and required next step	Proposed lead	Timing and output	Deep dives
S1 Situation	Demand baseline and regional market Have: provincial sales history and map. Reconcile six flagged years; refresh a complete year.	Manish Nigel review	W1–W2 Sourced demand baseline	3 / 4
S2 Situation	Domestic supply, finished imports and entry ports Have: 2024 sales/trade; diesel imports conflict. Gap: matched output/stocks and product-specific port allocation.	Manish Nigel access	W1–W3 Fuel balance + port bridge	5 / 6
C1 Complication	Demand drivers and levers Have: indexed passenger/freight stocks, agriculture/industry, power and road/rail. Source ICE splits and intensities; calibrate litres effects.	Manish Nigel review	W1 define; W2–W4 test Lever-to-demand bridge	7
C2 Complication	Refinery and plant scenarios Have: published capacity history and conditional CEF case. Agree timing/yields; resolve repeated keys and plant mechanisms.	Manish Henry / Nigel	W1–W3 Plant output + imports	8
C3 Complication	Economic accessibility and competing routes Cost map is illustrative. Compare Durban–NMPP–Lesedi, road/rail, Matola and Walvis Bay for the same product and destination.	Manish Nigel / Henry	W1 scope; W2–W4 test Delivered R/litre + access	9 / 10
R1 Resolution	Actual Vopak share and contestable customer demand Share is unknown. Test each candidate increment against cost, physical capacity, commercial rights and unique deliveries.	Nigel client data Manish analysis	W1 request; W2–W4 test Served + contestable litres	11 / 12 / 13
R2 Resolution	Storage and optional investment Partial published stocks are available. Verify usable product capacity; size additional service needs only after the incremental flow case.	Nigel priority Manish / Henry	W4–W6 if prioritised Flow-to-stock / investment	14 / 15

Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

Locate fuel demand alongside the infrastructure serving it

Petrol + diesel sales by province | 2022, bn litres/year



Evidence and implications

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Regions are working groups

Province boundaries define reporting groups; they do not establish terminal catchments.

03 | Sales are historical demand evidence

The map is sourced provincial sales, rather than a current-year outlook or commercial accessibility result.

Next steps | proposed owners

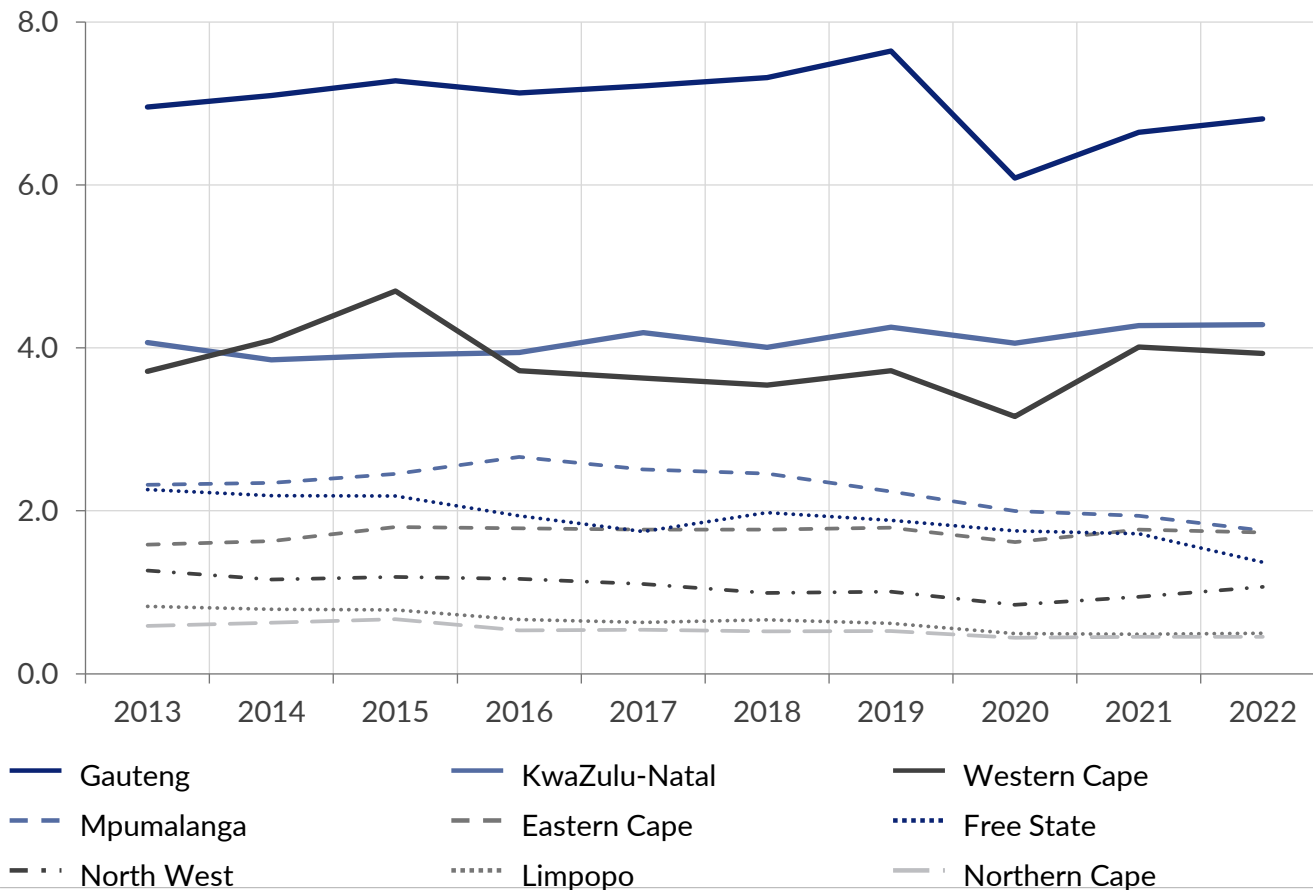
Manish: source a later complete year and test actual destinations, route costs and customer access.

Lease offers are not operating capacity. *Magdala RFP location narrative conflicts; briefing coordinate used. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

Track petrol and diesel sales by province over time

Annual provincial sales | 2013–2022, bn litres/year



Evidence and implications

01 | History ends in 2022

The latest complete provincial year in this extract is 2022.

02 | 2023 is incomplete

Only Q1 is present; it is excluded from annual comparisons.

03 | Six years are flagged

Provincial sums differ from national sales in 2013, 2014, 2015, 2017, 2018 and 2021.

Next steps | proposed owners

Manish: reconcile each flagged value to its original workbook cell and log old value, proposed value and reason.

Collect the national fuel balance and expose source disagreements

National petrol/diesel | 2024, billion litres

COLLECTED | trade and sales; production/stock reconciliation remains open

Product	Reported sales	Imports	Exports	Net imports
Petrol	9.029	4.000	0.920	3.080
Diesel	11.734	10.800	0.790	10.010
Combined	20.763	14.800	1.710	13.090

Source flag | 2024 diesel imports

FIASA staged: 14.793 | Government report: 10.800

Difference: -3.993 bn L. Preserve both; investigate scope and vintage.

Domestic production remains a dated observation

Latest staged balance: 2021 petrol 6.35; diesel 5.31 bn L.

Do not combine 2021 output with 2024 sales/trade to close a balance.

Evidence and implications

01 | Public figures are collectable

2024 petrol/diesel imports total 14.8 bn L in the government report. Exports total 1.71 bn L; these are rounded reported figures.

02 | The sources disagree

FIASA reports 14.793 bn L of diesel imports for 2024 versus 10.8 bn L in the government report. Neither value has been silently replaced.

03 | Port cargo has wider coverage

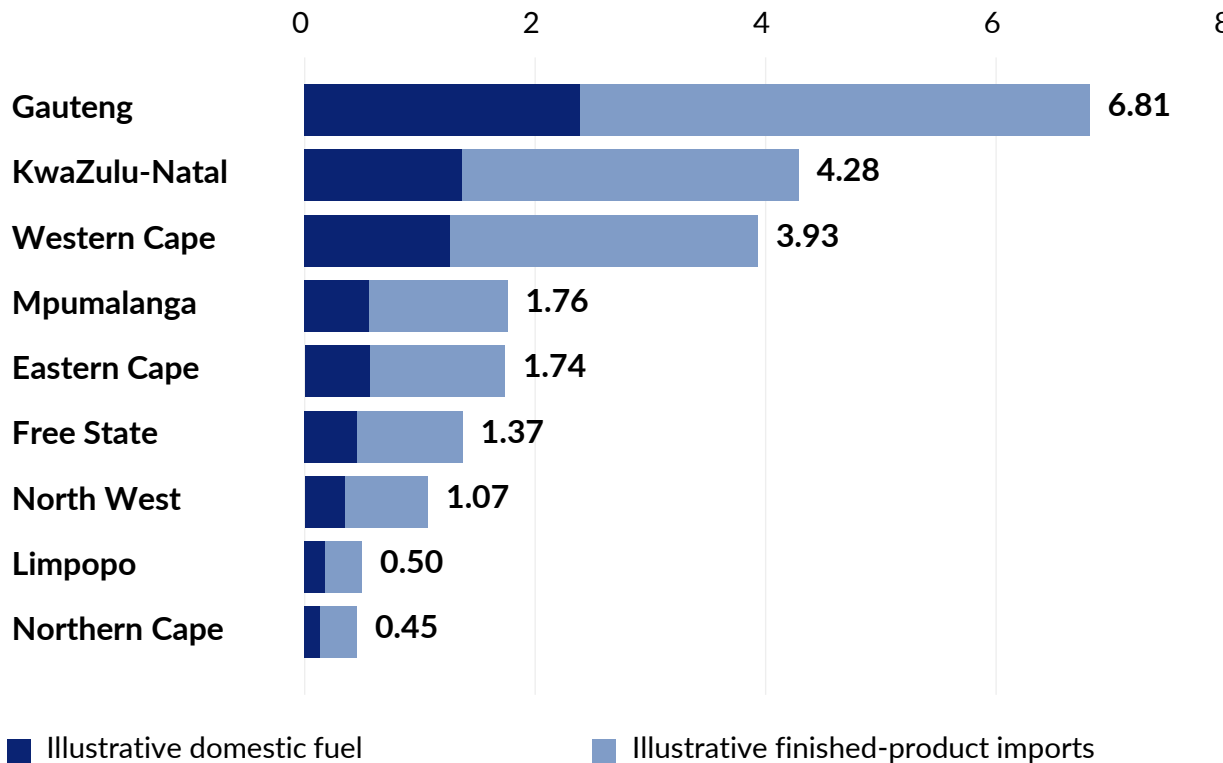
TNPA publishes liquid-bulk cargo by port, but it includes crude and other liquids. It does not establish petrol/diesel import allocation.

Next steps | proposed owners

Manish: reconcile raw SARS product codes and units, then source matched-year production and stocks. Nigel: confirm access to product-specific port data.

Shape the provincial split between domestic fuel and imports

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

Evidence and implications

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

Next steps | proposed owners

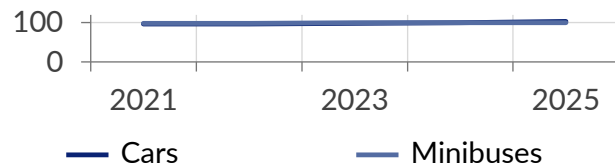
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Track passenger, freight and sector activity driving fuel demand

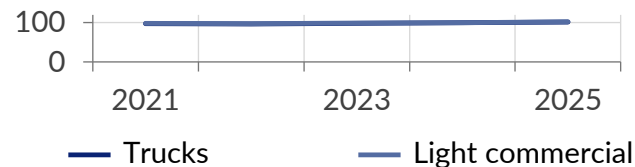
Demand-driver dashboard | every series indexed to 2024 = 100

Common 0–120 scale. Activity proxies; vehicle ICE split unresolved.

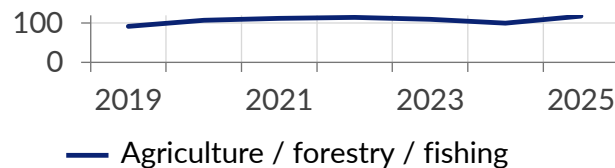
Passenger stock | all fuels; Dec



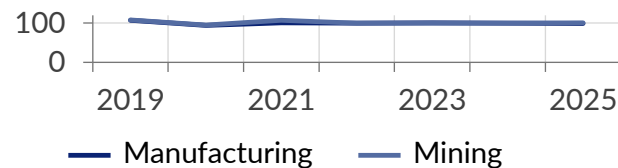
Freight stock | all fuels; Dec



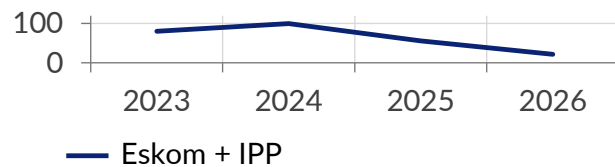
Agriculture | real value added, CY



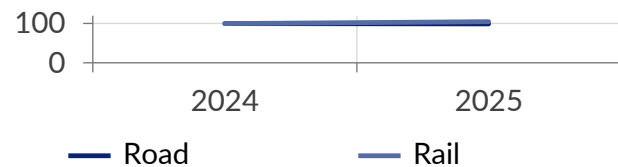
Industry | real value added, CY



Power | OCGT generation, FY



Freight activity | payload, CY



FY ends March; CY is calendar year. Stock is December snapshot. ICE = internal combustion engine.

Evidence and implications

01 | Passenger and freight ICE need a split

NaTIS separates cars/minibuses and trucks/light commercial, but not petrol, diesel or electric. Stocks are all fuels; kilometres and efficiency remain open.

02 | Sector activity is now explicit

2025 versus 2024: agriculture +17.4%; manufacturing -1.2%. Value added is an activity proxy, not direct fuel consumption.

03 | Uptake and modal levers remain distinct

2025 sales: 1,088 BEVs; 12,818 hybrids. Rail payload +4.7%; road -0.3%. Hybrids still use fuel; payload is not tonne-km.

Next steps | proposed owners

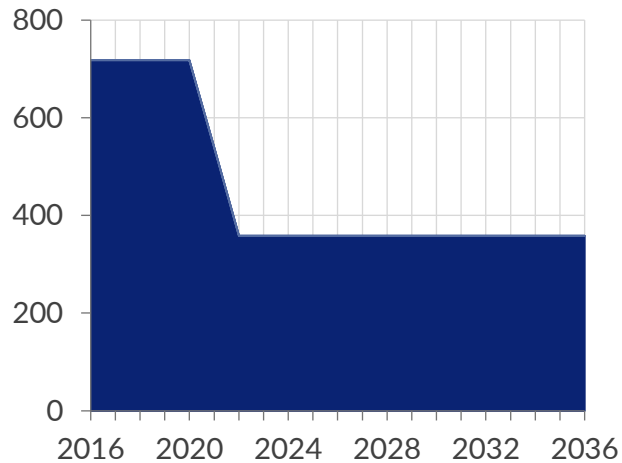
Manish: source ICE fuel splits, vehicle-km, freight tonne-km and sector fuel intensities. Keep the freight revision flag; refresh prices (annual extract ends 2023). Nigel: agree lever alternatives.

Compare the capacity history with a conditional redevelopment case

Published footprint + illustrative scenarios | thousand bbl/day

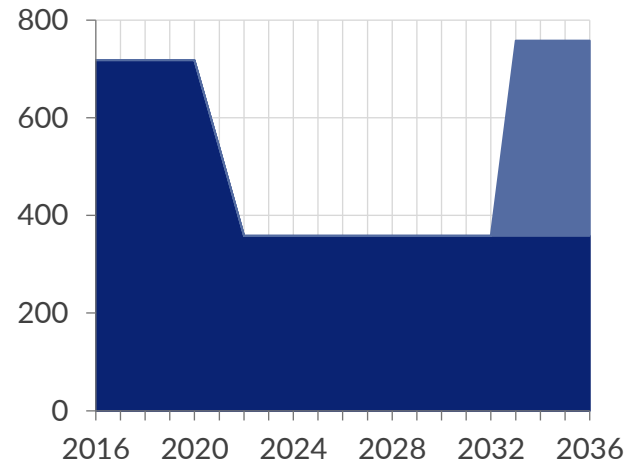
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

B adds 400 from 2033; darker navy = existing footprint; lighter blue = conditional addition.

B assumes illustrative FID 2029 + 48 months; no sanctioned date. Both hold existing capacity flat.

Evidence and implications

01 | History shows a structural decline

Published total falls from 718 in 2016–2020 to 538 in 2021 and 358 from 2022–2025. Capacity does not measure realised output.

02 | Redevelopment is conditional

CEF proposes approximately 400 thousand bbl/day after FID. Construction is approximately 48 months; the 2029 FID here is authored for comparison.

03 | Product output still needs modelling

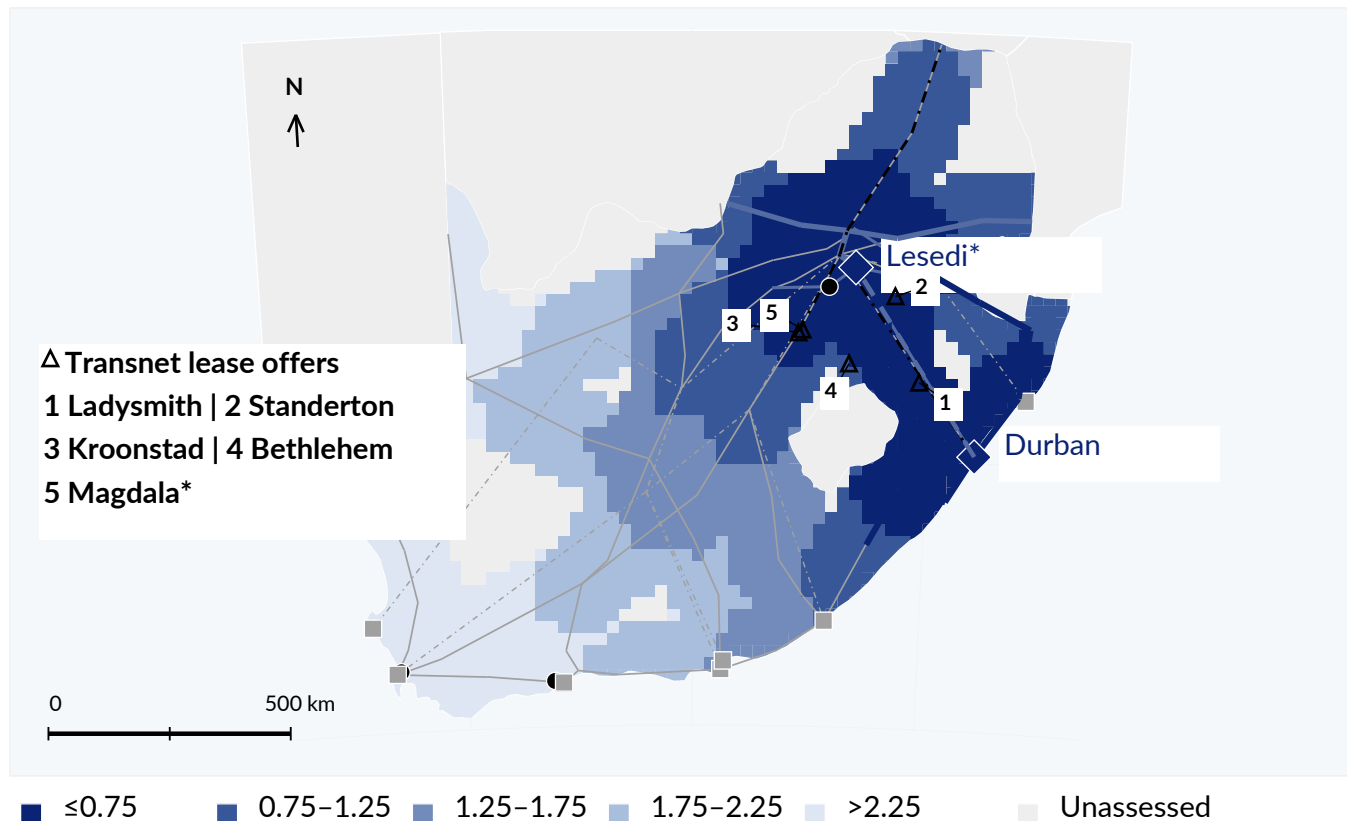
Source utilisation, downtime and product yields before translating capacity into domestic supply. Secunda/Natref cases remain open.

Next steps | proposed owners

Nigel: agree scenario timing. Manish: reconcile plant capacity and output/yields. Henry: review technical feasibility and the Sasol gas-to-liquids mechanism.

Map conditional market access from Durban and Lesedi

Road-delivery accessibility | illustrative transport cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

Evidence and implications

01 | Darker cells mean lower example cost

The surface shows illustrative delivered road-transport cost in R/litre.

02 | Other modes remain context

Pipeline and rail costs, capacity and access are not evaluated in this surface.

03 | Physical proximity is insufficient

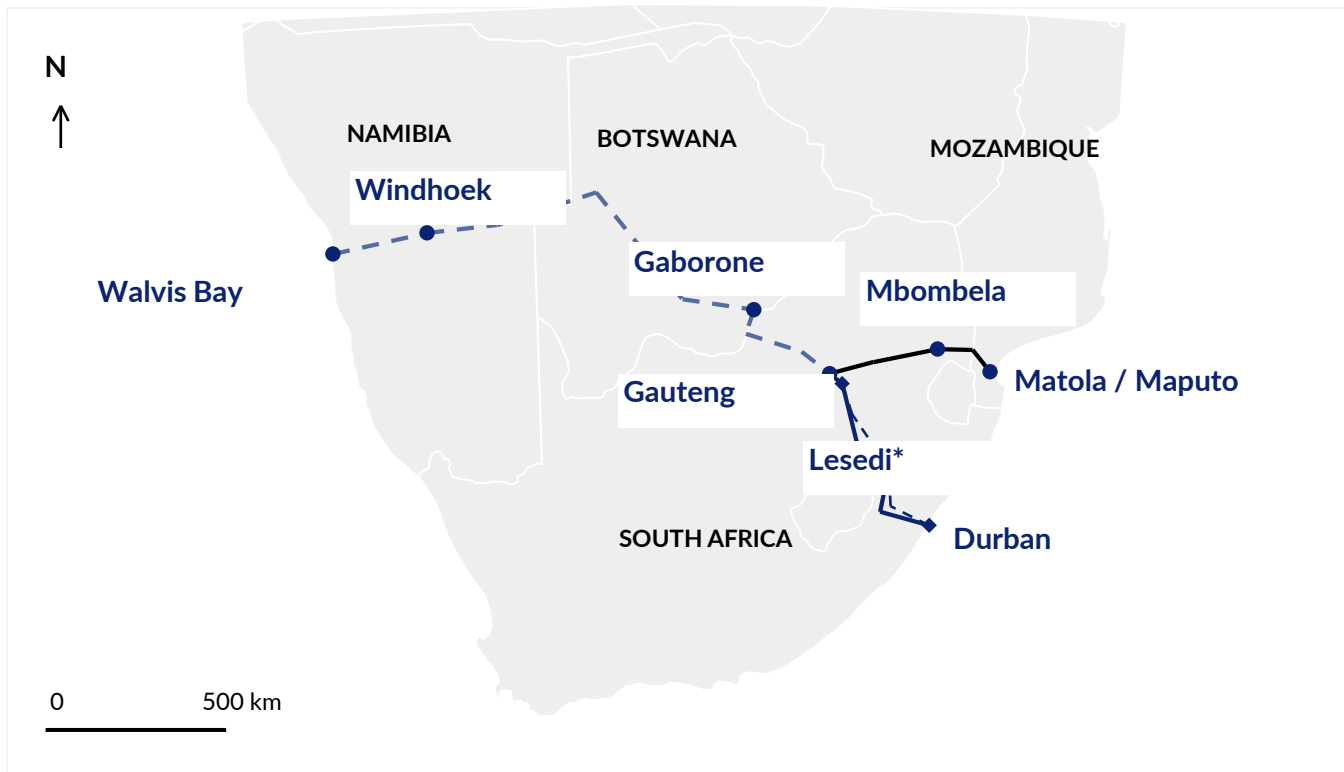
No operational or commercial catchment is validated by this illustration.

Next steps | proposed owners

Manish and Nigel: select the first destination and obtain dated route tariffs, quotations and access evidence.

Locate alternative gateways to the same inland customers

Durban, Matola and Walvis Bay | schematic corridor comparison



Solid: Durban / N4 candidates | Dashed: Trans-Kalahari / NMPP

LAEA / WGS84; routes schematic. No cost, available capacity, fuel rights or catchment is inferred.

Evidence and implications

01 | Compare the same inland destination

Gauteng is the common comparison market. Lesedi is a Vopak facility, not a destination every competing route must pass through.

02 | Matola approaches from the east

N4/Maputo connects Mozambique to Gauteng via Komatipoort and Mpumalanga. Fuel-compatible terminal access and road/rail service need testing.

03 | Walvis is an inland comparator

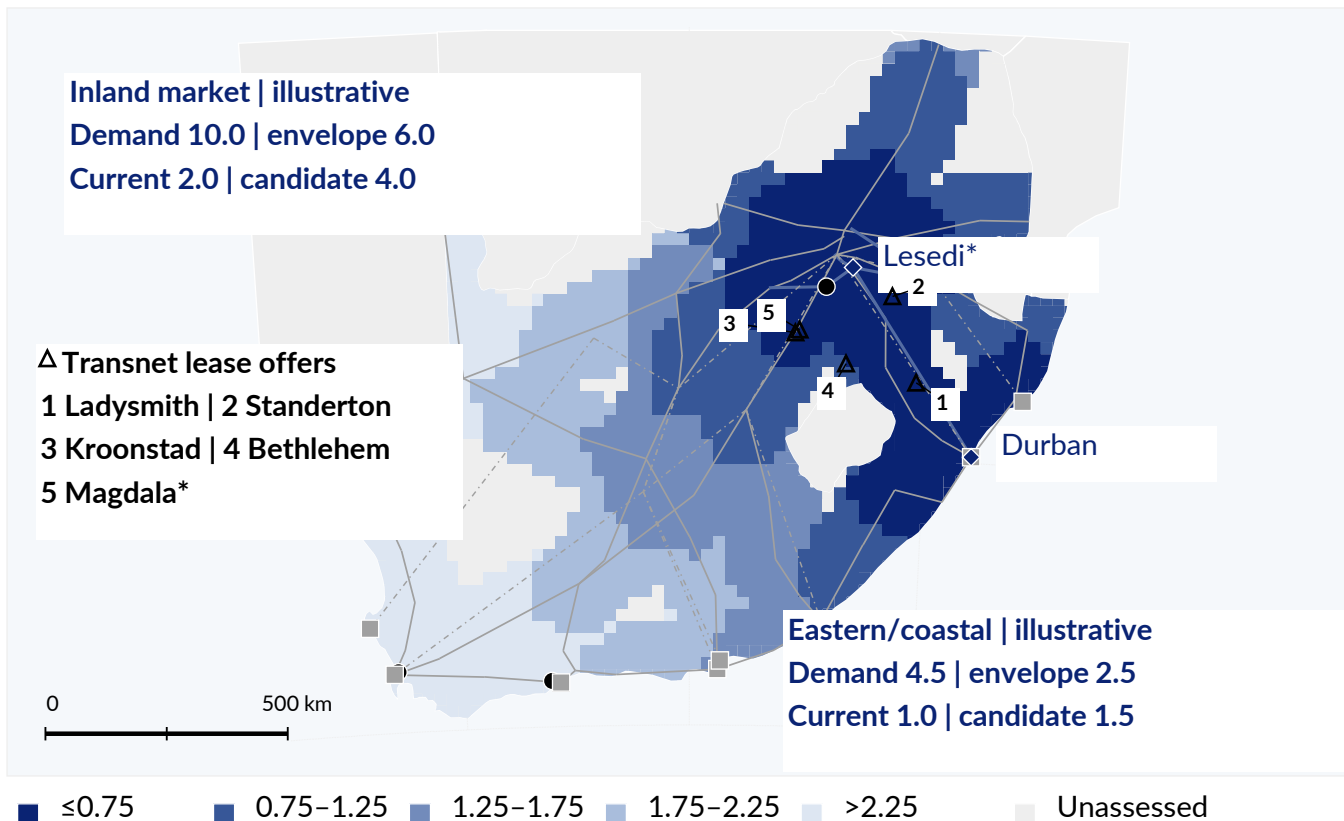
Trans-Kalahari links Walvis Bay via Namibia and Botswana to Gauteng. Its competitiveness for specific inland customers is unassessed.

Next steps | proposed owners

Nigel: select the customer destination. Manish: compare handling, transport, border/FX costs and access on the same R/litre basis; keep unevidenced routes unassessed.

Separate demand, accessible volume and Vopak flows

Illustrative road cost (R/litre) and market volumes



Receipts 2.8 + 2.0 – transfer 1.8 = unique demand 3.0

Example rates: R0.15/L + R0.002/L/km. Grey = unassessed. Volumes: bn L/year.

Source: Illustrative CSVs and road transport rates; no actual Vopak throughput or quoted transport costs supplied.

Evidence and implications

01 | Balance arithmetic is illustrative

Example demand 21.0 less domestic production 7.0 gives imports 14.0, before exports and stocks.

02 | Transfers count once

Example Durban receipts 2.8 plus Lesedi receipts 2.0 less shared transfer 1.8 gives unique served demand 3.0.

03 | Candidate volume is conditional

The example envelope is 8.5 and additional candidate volume 5.5; neither is forecast capture.

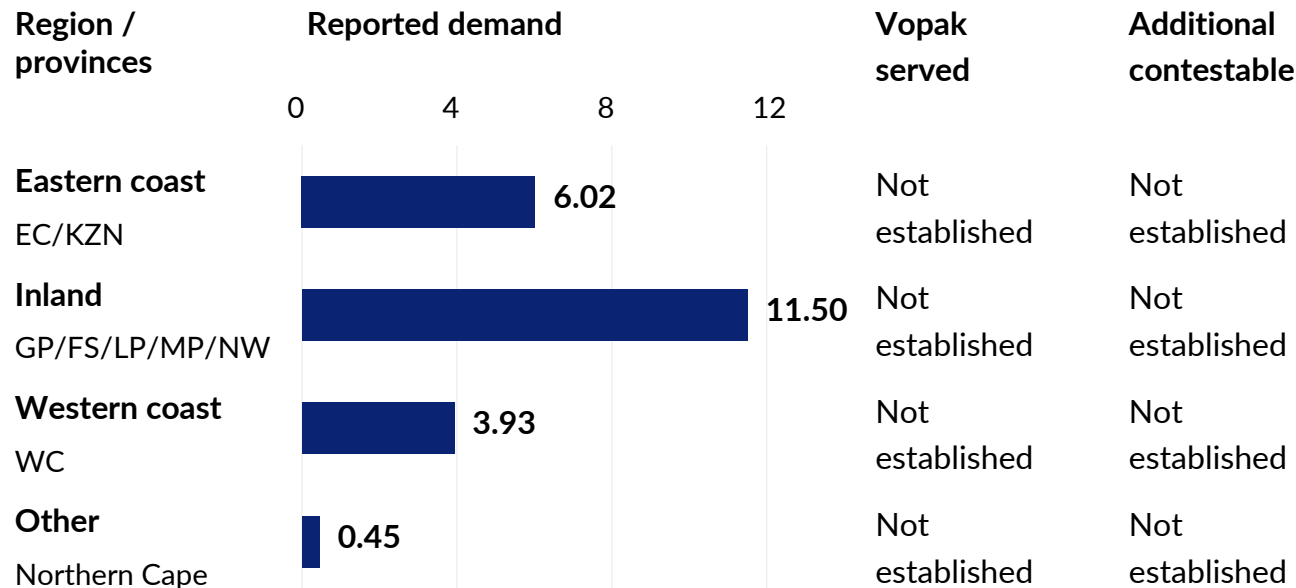
Next steps | proposed owners

Nigel: request client flows. Manish: replace examples with matched-year product and destination records.

Size each regional market and locate competing storage

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded



To calculate share: unique Vopak customer deliveries / same-year regional demand.

Contestable demand needs route cost, compatible tanks, contracts and switching evidence.
Count Durban-Lesedi transfers once. Region boundaries are working groupings, not catchments.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.
Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].
Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

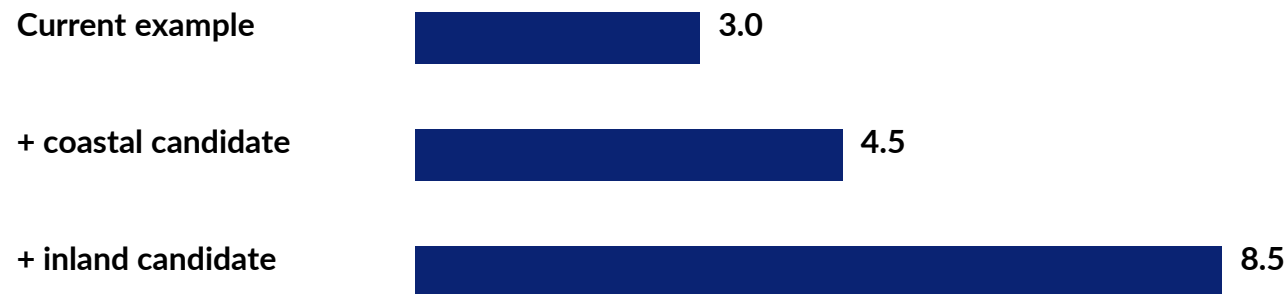
04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

Show what must hold to reach each candidate volume

Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts



What must be true for the additional volume?

Gate	Coastal +1.5	Inland +4.0
Delivered cost	≤ agreed customer threshold	Competitive full route R/litre
Physical access	Compatible receipt / dispatch	NMPP or alternative route capacity
Commercial rights	Customer access / switching	Destination / contract access
Volume accounting	Unique final deliveries	Remove shared Durban–Lesedi transfers

Evidence and implications

01 | Reach is necessary, not sufficient

Accessibility map p9 and competing routes p10 locate candidate access. Neither currently links a verified cost threshold to customer litres.

02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

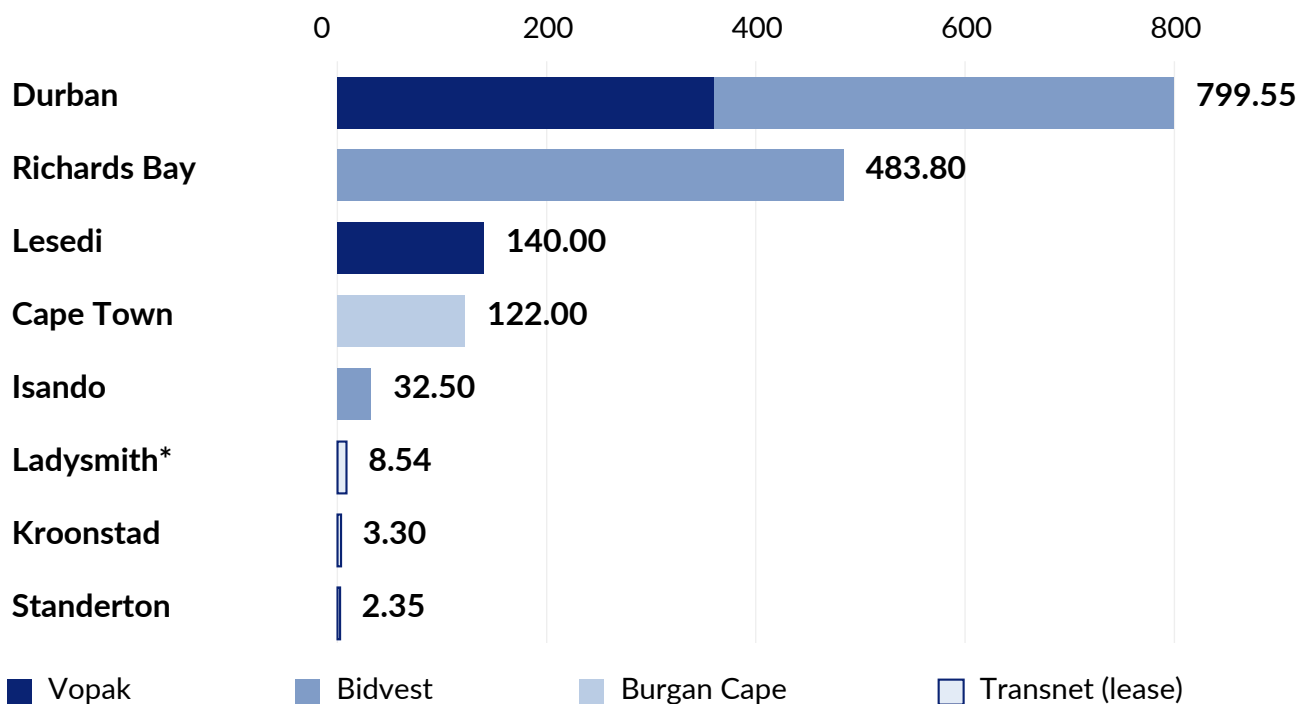
Next steps | proposed owners

Nigel: agree destinations and acceptable delivered cost. Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

Compare published storage capacity by location

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Borgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

Evidence and implications

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

Next steps | proposed owners

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Gate investment on evidenced incremental customer flows

Resolution | optional service and investment assessment

EVIDENCE GAP | analysis specification, not a populated result

New storage and investment scope

Incremental accessible flows, turnover/peak inventory, product compatibility, cost and commercial return criteria.

Output: Throughput-to-usable-stock bridge with seasonality, inventory days and dispatch limits; downside sensitivity; no investment claim from import deficit alone.

Evidence required before quantification

Partial gross/lease stocks, without client handling/inventory profile.

Detailed service sizing deferred; infrastructure secondary to fuel integration.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Partial gross/lease stocks, without client handling/inventory profile.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Nigel: decide whether to prioritise investment analysis after the flow case. Manish and Henry: bridge incremental flows to usable stock and service constraints.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

